KSBMB INTERNATIONAL CONFERENCE 2022
OVER THE PANDEMIC: ACADEMY, TECHNOLOGY & INDUSTRY
MAY 23MON-26THU, 2022
BEXCO, BUSAN, KOREA

[D. Cancer biology] D-132

Inhibitory effect on TC-1 lungcarcinoma cell proliferation and cell viability of Akebia quinata

Hyemi Yun[1], Dahyun Kim[1]*

[1]Biomedical laboratory science, Daegu Haany University, Daegu 38609, Korea

Lung cancer ranked first in the global cancer incidence rate for 20 years, and ranked first in the domestic cancer mortality rate in 2018. As such, the risk of lung cancer increases but there is no clear treatment for it. So there is a need for research on natural anticancer drugs. Akebia quinata, medicinal herb, is known to have anticancer effects such as breast cancer and liver cancer and to have antioxidant effects. In order to find out whether Akebia quinata is also effective for lung cancer, WST assay and Wound healing assay were performed. The results of measuring the anticancer effect by WST assay of Akebia quinata extract, a concentration dependent decrease in cell viability was observed. The wound healing assay, an experiment to observe hyperproliferation and metastasis of cancer cell, was performed. It was confirmed that the gap narrowing time obviously increased as the concentration increased. Through the above results, it can be seen that the Akebia quinata has the effect of inhibiting cancer cell survival rate, hyperproliferation and metastasis. As a result, It is expected that Akebia quinata could be used as a potential source of natural anticancer drugs at lung cancer.

Inhibitory effect on TC-1 lungcarcinoma cell proliferation and cell viability of *Akebia quinata*

Hye-mi Yun[1], Da-Hyun Kim[1]

[1]Biomedical laboratory science, Daegu Hanny University, Republic of Korea

ABSTRACT

This research was conducted to investigate the anticancer effect and mechanism of extracts of *Akebia quinata(Aq)* against TC-1 cells, a lung cancer cell line. The stems and leaves of *Aq* powder was extracted with DMSO. To determine whether extracts of *Aq* were effective in other cancers such as melanoma, gastric cancer, ovarian cancer, and lung cancer, we investigated the expression of *p53* and *RB* in cell lines of each cancer (B16F10, AGS, OVCAR-3, and TC-1 respectively) by Westernblot and RT-PCR. As a result, a distinct increase in the expression of the above genes and proteins was observed in TC-1. The results of measuring the anticancer effect by WST assay of *Aq* extracts, a concentration dependent decrease in cell proliferation was observed. As a result of the time-dependent WST assay at a concentration of 40 ug/ml, it was observed that the cell proliferation decreased with time. qPCR was performed to quantitatively investigate which anti-cancer mechanism the *Aq* extracts kills cancer cells. As a result, as the treatment time of the extracts increased, the expression levels of *p53* and *RB* genes, which are cancer suppressors, and *p21* and *p27* genes, which are cell cycle arrest factors, increased. In addition, it was confirmed that the expression levels of *TGF-β*, *TGF-α*, and *F4/80* genes, which are inflammation-related factors, decreased with time. Through this, it was found that the extracts of *Aq* extracts showed anticancer effect by inducing apoptosis and cell cycle arrest of TC-1 cells. Therefore, it is considered that the extracts of *Aq* has potential as a candidate material for the treatment of lung cancer.

INTRODUCTION

Lung cancer refers to abnormal cancer cells in the lungs that proliferate uncontrollably to form a mass and harm the human body. However, when lung cancer progresses, it can metastasize not only to the contralateral lung but also to the body through lymph nodes or blood, such as bones, liver, adrenal glands, kidneys, brain, and spinal cord. According to the International Agency for Research on Cancer, lung cancer is the most common cancer in the world, and lung cancer is estimated to be the number one cause of cancer deaths. As such, the risk of lung cancer increases, but there is no clear treatment for it. So there is a need for research on natural anticancer drugs. *Akebia quinate(Aq)* is a deciduous plant of the *Akebia quinata* family, distributed in Korea, Japan, and China. *Aq*, medicinal herb, is known to have anticancer effects such as breast cancer and liver cancer and to have antioxidant effects. In this study, we investigated the anti-cancer effect and anti-cancer mechanism of extracts from the leaves and stems of *Aq* against TC-1, a lung cancer cell line.

RESULT

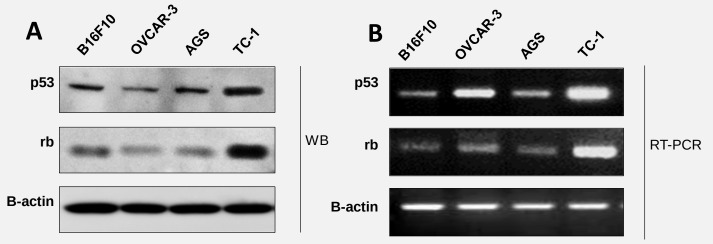

Figure 1: Expression of oncogenes in TC-1 cells treated with *Aq* extracts
(A) Expression of *p53* and *RB* proteins, cancer suppressor factors, in TC-1 cells treated with *Aq* extracts by Westernblot. *B-actin* was used as a loading control.
(B) Expression of *p53* and *RB* genes, cancer suppressor factors, in TC-1 cells treated with *Aq* extracts by RT-PCR. *B-actin* was used as a loading control.

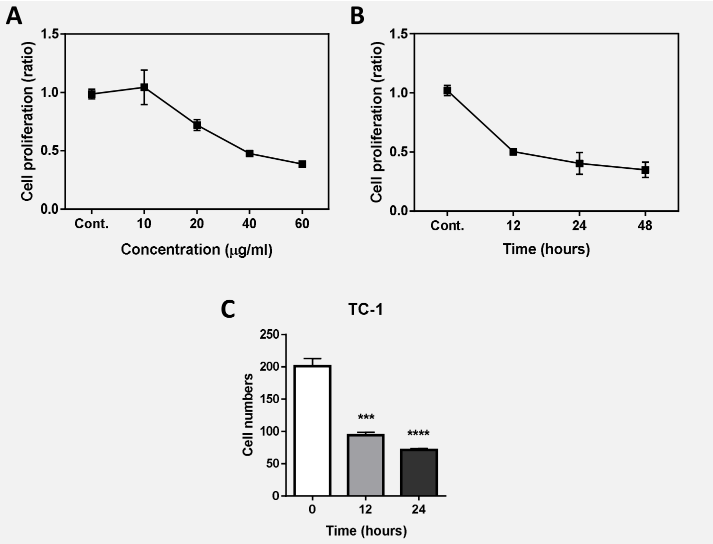

Figure 2: Inhibition of cell viability and proliferation in TC-1 cells treated with *Aq* extracts
(A) Concentration dependent decrease of cell proliferation in TC-1 cells treated with *Aq* extracts by WST assay.
(B) Time dependent decrease of cell proliferation in TC-1 cells treated with *Aq* extracts by WST assay.
(C) Decrease of TC-1 cell number with time.

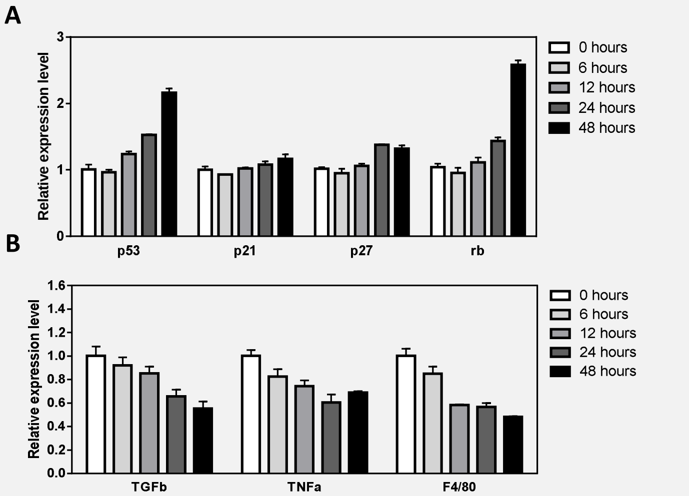

Figure 3: Quantitative changes in TC-1 cell oncogenes
(A) The expression levels of *p53* and *RB* genes, which are cancer suppressors, and *p21* and *p27* genes, which are cell cycle arrest factors, increased with time by qPCR.
(B) The expression levels of *TGF-β*, *TGF-α*, and *F4/80* genes, which are inflammation-related factors, decreased with time by qPCR.

CONCLUSION

- Expression of cancer suppressors *p53* and *RB* in TC-1 cells was increased by treating *Aq* extracts.
- *Aq* extracts treatment inhibits TC-1 cell proliferation and cell viability concentration and time dependently.
- *Aq* extracts kill TC-1 cells by the mechanism of activating apoptosis and cell cycle arrest.
- *Aq* is considered as a natural anticancer candidate for lung cancer.